Smart Lab - final execution report (closure and optimisation mission)

Work done 10-11 November 2026 on the owner's laptop (Dell G3 3590). The file name keeps the date the mission was issued. Prepared for the owner (Ali Loay) and thesis supervision.

Status words used throughout: **LIVE** = deployed and checked on the running system today; **TESTED OFFLINE** = automated tests on this laptop; **MEASURED** = numbers taken on this laptop today; **LAB** = needs the real lab, people and cameras (see the physical acceptance checklist); **OPEN** = not done.

1. Executive summary

- **Deployed and live:** all portal and backend changes (security fixes, faster tabs, streamed AI answers, read caches, live camera activity bar, AR multi-PC boxes). Camera services restarted on the new code.
- **Cameras:** a person the cameras named before comes back as a marked guess "Name ? (look)" after leaving the view or after a service restart, and is confirmed by the face; no name is ever given from a head count alone; the look handoff between the two cameras is in. **Simulation-tested only** - the two-person walk test is LAB.
- **Speed:** revisited portal pages paint at once from the last answer; AI text appears while it is written; the dashboard summary and lab catalogue are shared for 2-3 s; instant AI answers now include what the cameras see.
- **Laptop:** the bottleneck is memory, not CPU. Camera CPU work cut by ~0.5 core, ~1.5 GB of AI models no longer loaded at launch. **C: free space 15.2 GB -> 85.8 GB** (training data and Docker's disk moved to D:, project moved out of OneDrive to C:\Projects\smart-lab-portal).
- **Hardware:** no second computer is justified. If money is spent: 2 x 16 GB RAM (~3,250 EGP).
- **Not pushed, not committed** (owner decision). No door, firmware or robot action was taken.

2. Repository and system state

Item	Before	After
Project folder	C:\Users\dell\OneDrive\Desktop\Bsc\codes\smart-lab-portal	**C:\Projects\smart-lab-portal** (OneDrive copy removed; empty folder left while VS Code had it open)
Branch / commit	qa/browser-and-camera-docs @ fdfd1fe	same branch and commit; 58 files changed and 16 new files, uncommitted
Docker disk	C:\...\Docker\wsl\disk\docker_data.vhdx (67.6 GB)	**D:\DockerData** (verified: same 5 volumes, same row counts)
Training data	tools\vision_train\data (11 GB on C:)	**D:\SmartLabData\vision_train_data**, the old path is a link
C: free	15.2 GB	**85.8 GB**
Portal images	built 10 Nov	rebuilt 11 Nov; rollback tags `smart-lab-portal-{backend,frontend}:rollback-20261111`

Backups (D:\SmartLabBackups\2026-11-11_predeploy): database dump (43 tables), uploads and n8n volume archives, the whole git repository with every branch (bundle), the uncommitted diff, Docker settings, row counts. **Full project copy:** D:\SmartLabBackups\project_copy_2026-11-11 (94,096 files, byte-identical). Git recovery branch: `backup/before-closure-2026-11-11`.

3. Sources and how they were used

25 Argon files in Downloads (35 distinct files by SHA-256 among 45 similar names) and the two "Master Unified Engineering Execution Blueprint" uploads (identical, SHA-256 6957fbfe...). The Argon patch `Smart_Lab_Optimization_Fixes_2026-11-10.patch` was **not applied**: it targets an older snapshot. Each finding was checked against today's code.

Argon / blueprint finding	In today's code	Decision
SEC-01: five door routes without device scope	confirmed	fixed (stricter than proposed: door devices only)
SEC-02: PC key can report as another kind	confirmed on /equipment/state	fixed
SEC-03: PC enrolment open by default	confirmed, and compose could not close it	fixed
SEC-04: token in WebSocket URL	confirmed	fixed with one-time tickets
SEC-05: /scan/appearance unthrottled	confirmed	fixed
SEC-06: issue text in AI prompt	already treated as data in the system prompt; staff-only	kept; no XML-only "fix"
P1-A: cross-camera naming by elimination	confirmed	fixed: needs a measured agreeing look
Lower MIN_DEEP_H 80 -> 48, wider floor gates, 2-pass warm-up	no evidence it is safe	rejected until measured on lab footage
"tools/camera/..." paths	real code is tools/security_vision	used the real paths
Unify on qwen2.5:1.5b with 24 h keep-alive	live already uses 1.5b; 24 h holds RAM	keep-alive not raised (RAM is the bottleneck)
AR: skip duplicate detector, 4 threads	the bin model is already skipped when the lab model loads	multi-instance boxes added instead
Composite DB indexes	no slow query measured	not added (no evidence)
Watchdog with auto-restart	camera restarts leave gaps	monitor-only watchdog

4. Every change

4.1 Security (TESTED OFFLINE, LIVE)

Fix	Files	Tests
Door reports accept a device's own key only for itself and only from door devices (master, door camera, face server); the shared door key is unchanged	backend/app/api/routes/access.py	test_device_keys.py +2
A device's own key must report as its own kind (a PC key cannot file a robot report)	routes/equipment.py	test_pc_keys.py +1
PC enrolment closed by default; `PC_ENROLL_OPEN` and `DEVICE_SHARED_KEY` passed through docker-compose; start-up warnings for open enrolment / open sign-up	core/config.py, main.py, docker-compose.yml	existing + conftest
Live-updates socket uses a 30 s one-time ticket (`POST /api/auth/ws-ticket`) instead of the sign-in token in the URL; nginx access log without query strings	core/ws_ticket.py, routes/auth.py, main.py, lib/live.tsx, nginx.conf, app_locations.conf	test_portal.py +1

Fix	Files	Tests
Phone box-match limited to 6 frames / 2 s per user	core/throttle.py, routes/scan.py	test_appearance.py +1

Owner action: set `PC_ENROLL_OPEN=true` in `.env` only while installing lab PCs.

4.2 GitHub CI failure (TESTED OFFLINE)

Main's CI failed on `test_qr_pipeline` (a random booking token decoded at 2 of 7 distances). Measured locally: 392 of 400 random tokens decode at 7/7 distances, 7 at 6/7, 1 at 5/7. `issue_token()` now also checks each token through the door-camera model and draws again below 6/7, so a weak token is never issued (qr.py; 2 new tests).

4.3 Cameras and identity (TESTED OFFLINE; LAB for real walks)

- **Name kept after leaving / restart ("look guess"):** a person the camera named before (open door entry, or a face name from the last 15 min - both kept on disk across restarts) who is back in view without a name gets it from the stored look as a **marked guess**, shown "Ali ? (look)" and "? guess by look" on the Security page. Rules: clear look (<= 0.18), clearly closer than every other stored person and every other unnamed person (margin 0.10), alone, upright, big enough, held ~2 s. It is never shared to the other camera, never learnt from, yields to a new door entry, is dropped if this camera's face says someone else, and becomes a real name only when the face agrees or the other camera's confirmed name is on the same spot. Switch: `SMARTLAB_LOOK_GUESS` = on / observe / off. Files: vision.py, identity.py, crosscam.py; test_look_guess.py (9 tests).
- **Door entry -> instant name:** already in place (`Labeler._door_entries`): the person who walks in right after a granted entry is named; two people on one entry stay unnamed with a review note. Unchanged.
- **No naming by head count:** cross-camera "elimination" now requires a measured look that agrees; otherwise one `CROSSCAM_AMBIGUOUS` note. test_crosscam.py (+2, 1 rewritten).
- **Look handoff between cameras** (local commit 68b7157) brought in: a confirmed name crosses on a clear look when feet are hidden or the person already left the other view (12 simulated tests).
- **Live activity bar:** one line per tracked person - name or "Person N", upper-body colour, posture (standing, walking, seated or leaning, bending or crouching, lying), and the mapped box they are at or near ("most likely at UR5e"). Built from data the service already has: no new model, no extra GPU work. Shown as a scrolling LIVE bar under each lab's cameras and in the assistant's instant answer. Files: activity.py (new), security_vision.py, routes/security.py, Security.tsx, agent.py; tests test_activity.py (4), test_camera_activity.py (2).
- **Head/upper-body tracking:** the full-person detector already boxes partial people (upper body, seated, behind desks: YOLO26s 544x960 + zoom pass + low-confidence second stage). A separate head detector or pose model was **not** added: it would add GPU load to both cameras and no measurement shows a gain. The name logic above uses appearance and face, so it does not need the feet.
- **Half-body looks (MEASURED):** a box cut by the bottom of the picture is now a waist-up view: its crop fills only the top of the re-id model's input and its colours are compared shirt-to-shirt. On two lab recordings (348 same-person pairs) the same-person distance between a full-body and a waist-up view fell from p50 0.282 / p95 0.561 to **p50 0.170 / p95 0.453**; other people stayed at 0.59-0.66 (only 5 pairs - to be confirmed with the two-person lab recording). Waist-up views never update a stored identity. Switch: `SMARTLAB_PARTIAL_LOOK=off`. vision.py; test_partial_look.py (4).
- **Honest limits:** appearance cannot guarantee the same person across a change of clothes, and similar clothes can look alike; that is why the result is a guess until the face confirms it. Measured re-ID distances and the partial-body change are LAB items (section 9).

Scenario bench (simulated, 14 scenarios incl. crossings, posture, 1.5 s / 4 s occlusion, camera off 10 s, two cameras): **0 false names, 0 name switches, 0 ID switches.**

4.4 Portal speed (TESTED OFFLINE, LIVE)

- **Last answer first:** `api.peek()` returns the last answer of the same read for this sign-in (max 60 s, cleared by any write and by sign-out, never for alarms, emergency, sharing or sign-in). 9 pages paint from it at once and reload in the background: Admin / Staff / Student dashboards, dashboard widgets, Labs, Devices, My bookings, Alerts, Equipment, Users.
- **Backend read caches:** `/api/summary` 3 s and `/api/labs/overview` 2 s shared caches; a signed-in user's write empties them before and after the write (`ClearOnWrite`), so a change shows on the next read. Session expiry scan at most every 3 s instead of on a dozen GETs.

4.5 AI assistant (TESTED OFFLINE, LIVE)

- **Streaming:** `POST /api/ai/ask/stream` and `/api/ai/student/ask/stream` (one JSON object per line: meta, token, final or error). Same role checks, hourly limit and audit as `/ask`, which is unchanged. The chat shows the text as it is written; nginx no longer buffers /api/ai/. The one-answer-at-a-time slot is kept (cameras first).
- **Instant answers:** "what is everyone doing / who is doing what" now answers from the door records plus the cameras' live lines, without the model.
- **Measured:** answer model cold 8.0 s vs warm 0.4 s to a short reply; embedding model cold 1.1 s.
- **Maintenance reports through the assistant:** "the oscilloscope in LAB_01 is broken" / "report a network problem on PC-04" -> a proposal with lab, device, category and severity; **Confirm** files it through the normal `POST /api/issues` as the signed-in user (same checks and audit). Questions ("which issues are open?") are not mistaken for reports; booking sentences still go to bookings. agent_actions.py, ai.tsx; test_agent_issue_action.py (4).
- **Acknowledging alerts through the assistant (staff / admins):** "acknowledge the alerts", "ack alert 12", "acknowledge the camera alert" -> up to 5 open alerts, most severe first; **Confirm** calls the Incidents page's own `POST /api/incidents/{id}/actions` (staff only, audited, refused if already acknowledged). Students are told they cannot. test_agent_ack_action.py (3).
- Not built: image generation (no local image model is installed). Weekly / monthly reports, PDF and Excel exports already exist (Reports page, `/ai/report`).

4.6 AR scanner (TESTED OFFLINE, LIVE)

Per-class `max_instances` in detectors.json: lab PCs up to 6 boxes per frame; overlapping boxes stay one object; registered assets stay one box. Router, switch and master node were already reported together (test exists). **Colab round 3 package:** `D:\SmartLabData\colab\lab_r1.zip` (619 MB, 3,864 files) and `tools/vision_train/colab_train_lab_r3.ipynb` - trains YOLO11n, YOLO11s and an augmented YOLO11n on the same data, scores them exactly like round 1, exports the best to ONNX at 640.

4.7 Monitoring (TESTED OFFLINE, LIVE)

`tools/ops/watchdog.py` - checks 11 services; `--once` for a snapshot, otherwise logs only state changes to logs/watchdog.jsonl. Restarts nothing by design (a test checks that).

4.8 Resource changes (MEASURED)

Change	Evidence
Camera OpenCV threads 8 -> 2 (`--cv-threads`, `SV_CV_THREADS`); detector CPU fallback keeps 8	two lab recordings replayed at once

Change	Evidence
Annotated picture drawn 10/s only while someone watches, else 1/s (`SV_IDLE_RENDER_FPS`)	same
Knowledge service warm-up opt-in (`KB_WARMUP=1`)	1.5 GB held for 30 min after every launch

5. Performance measurements

Machine: i7-9750H 6 cores / 12 threads, 2 x 8 GB DDR4-2666 (both slots used), GTX 1650 4 GB, Windows Balanced.

At launch with the lab services up: free RAM 1.9 GB of 15.8, commit 28.8 of 42.8 GB (heavy paging). Docker's VM 2.3 GB (capped at 6 GB), local AI models 1.5 GB on the CPU, each camera service ~0.4 GB. Largest CPU user at rest: **OneDrive (~17 % of the machine)** syncing the project - fixed by moving the project.

Camera CPU work per service (2304 x 1296, 25 fps replay)	CPU cores	render p50 / p95
8 threads, render 10/s (before)	0.65	20 / 30-33 ms
2 threads, render 10/s	0.60-0.62	19-20 / 27-28 ms
2 threads, render 1/s when unwatched (now)	**0.41**	19-20 / 25-26 ms

Not measured today (LAB or later): live analysis rate with both cameras (target ~9/s each), AI time-to-first-word with cameras running, load test with 50-100 users, portal navigation timings in a browser.

6. Tests

See SMART_LAB_FABLE_5_2_FINAL_TEST_RESULTS.md. Summary: **backend full suite 775 passed, 2 skipped, 0 failed** (12 min on an in-memory test database); camera 328 passed; knowledge 49, PC agent 40, robot monitor 43, UR5e control 20, voice 6, LAN presence 6, relay 5, OPC UA 3, watchdog 3 - all passed; frontend type check 0 errors, build OK, node tests 15/15 (one test brought in line with the 3 Nov PC-status rule); MCP end-to-end not run (pytest missing in its environment).

7. Hardware decision

Software optimisation helps; the measured bottleneck is memory. No second computer is justified.

Option	Price found in Egypt (11 Nov, listings, not quotes)	Verdict
Software only (done)	0 EGP	about 1.5 GB RAM and 0.5 core back; OneDrive load gone
RAM 2 x 16 GB DDR4-2666 SODIMM	Crucial 16 GB at Technology Valley EGP 1,625 each, listed in stock: **~3,250 EGP**	fixes the bottleneck directly; check the G3 3590's 32 GB maximum first
Used mini PC 16 GB	Noon renewed from ~EGP 10,600; Dubizzle asking 6,750-9,250	2-3x the cost, a second machine to maintain
Raspberry Pi 5 8 GB	Fluxelectronix EGP 7,500; Makers Electronics EGP 14,000 (out of stock)	cannot run the camera AI path

8. Readiness

For	Ready?	Evidence
Local software demonstration	**Yes**	all suites pass, portal live and healthy, all 11 services up
Live laboratory demonstration	**Yes, after the checklist**	camera behaviour is simulation-tested; walk test pending
Final thesis defence	**Yes for software**; cite camera identity results as simulated until the walk test	
Deployment to another institution	**Not yet**	single-laptop setup, hand-set camera calibration, no install-from-scratch test on another PC today

4.9 Developer speed (TESTED OFFLINE)

With Docker's disk on the D: hard disk the backend suite would have taken about 5 hours. `tools/ops/test_backend_fast.ps1` runs it against a separate throw-away PostgreSQL held in memory: **7 min 44 s** for 772 tests. Live reads are not slowed by the move (API health with a database ping ~6 ms; queries ~6 ms, served from memory); bulk writes are.

9. Still open

1. **LAB:** two-person walk test, camera rates, restart test, AR on a phone (checklist).
2. **Owner:** commit and push (nothing pushed); run the Colab round 3 and bring back best.onnx + test_report.json.
3. **Software, next:** confirm the half-body result and tune thresholds on the two-person lab recording; AI model A/B with both cameras live; browser timing run of every tab; load test on an isolated copy.
4. **Optional purchase:** 2 x 16 GB RAM.

10. Commands

```
cd C:\Projects\smart-lab-portal
launch.bat                                        start everything (only what is missing)
stop_all.bat                                      stop everything (data kept)
py tools\ops\watchdog.py --once                   health of all services
py -m pytest -q tools\security_vision             camera tests
docker compose run --rm --no-deps -T -v "%CD%\backend:/app" backend python -m pytest -q -p
no:cacheprovider
```

Roll back:

```
portal images:  docker tag smart-lab-portal-backend:rollback-20261111 smart-lab-portal-backend:latest
                docker tag smart-lab-portal-frontend:rollback-20261111 smart-lab-portal-frontend:latest
                docker compose up -d --no-deps backend frontend
camera / AI switches (set before launch): SV_CV_THREADS=8  SV_IDLE_RENDER_FPS=15  KB_WARMUP=1
SMARTLAB_LOOK_GUESS=off
database:       pg_restore from D:\SmartLabBackups\2026-11-11_predeploy\smartlab.dump (insert-only merge,
never a wipe)
Docker disk:    restore D:\SmartLabBackups\2026-11-11_predeploy\docker-settings-store.json.bak
```

Smart Lab - final test results (11 November 2026)

All runs on the owner's laptop, from `C:\Projects\smart-lab-portal`, branch `qa/browser-and-camera-docs` with the uncommitted closure changes. Counts are per suite; they are not added together.

Suite	Command	Result
Backend, full	`tools\ops\test_backend_fast.ps1` (in-memory test DB)	**775 passed, 2 skipped, 0 failed** - 12 min
Backend, full (earlier, live server's test DB, before the last 3 changes)	old docker compose command	757 passed, 2 skipped, 1 failed: `test_robots::...disagreement_after_hold`; passed in the targeted re-run (121 passed) - intermittent timing
Security vision (cameras)	`py -m pytest -q tools/security_vision`	**328 passed**
Camera scenario bench (simulated, 14 scenarios)	`py tools/security_vision/scenario_bench.py`	**0 false names, 0 name switches, 0 ID switches**
Knowledge	`py -m pytest -q tools/knowledge`	49 passed
PC agent	`py -m pytest -q tools/pc_agent`	40 passed
Robot monitor	`py -m pytest -q tools/robot_monitor`	43 passed
UR5e control (dry run)	`py -m pytest -q tools/ur5e_control`	20 passed
Voice	`py -m pytest -q tools/voice`	6 passed
LAN presence	`py -m pytest -q tools/lan_presence`	6 passed
Relay	`py -m pytest -q tools/relay`	5 passed
OPC UA	Python 3.12: `-m pytest -q tools/opcua`	3 passed
Watchdog	`py -m pytest -q tools/ops`	3 passed
MCP end-to-end	-	**not run**: pytest is not installed in tools/mcp_server/.venv (needs the owner's OK to install)
Frontend type check	`npx tsc --noEmit -p frontend`	0 errors
Frontend build	`npx vite build`	built (chunk-size warning only)
Frontend node tests	`node --test e2e/*.test.mjs` (3 files)	**15 / 15** - one test updated to the 3 Nov PC-status rule (a live agent wins over the LAN scan)
Compose file	`docker compose config -q`	valid

Live checks after deployment (11 Nov)

Check	Result
`/api/health`	ok, database true

Check	Result
New endpoints exist (`/auth/ws-ticket` , `/ai/ask/stream` , `/ai/student/ask/stream`)	401 without sign-in (present, protected)
Door report without a key (`/access/grant`)	401
Served frontend contains the new code	ticket socket, streamed answers, live camera bar, "guess by look", create_issue
Watchdog snapshot	11 / 11 services up
Docker disk move	same 5 volumes; row counts identical (users 8, access events 2,005, audit logs 16,756, devices 26)
Project move	94,096 files, 11,741,328,078 bytes identical in old, new and D: backup

Not tested (needs the lab)

Real two-person walks with both cameras, camera analysis rate with both live, AR on a phone, assistant time-to-first-word with the cameras running, load test with 50-100 users. See SMART_LAB_FINAL_PHYSICAL_ACCEPTANCE_CHECKLIST.md.

Smart Lab - physical acceptance checklist (after the 2026-11-11 closure work)

Everything here needs the real lab, real people and the real cameras. Software tests cannot sign these off. Tick a line only with a measured result written next to it. Never test by unlocking the door from software, never move a robot from the portal.

Before you start

- ☐ `launch.bat` from **C:\Projects\smart-lab-portal** (the project left OneDrive on 11 Nov).
- ☐ `py tools\ops\watchdog.py --once` -> every required service UP.
- ☐ Both camera windows / logs show `re-id on directml-gpu...` and `OpenCV threads 2` behaviour (no errors).

Cameras and identity (two people: A enrolled, B enrolled)

- ☐ A enters through the door (fingerprint or face). Camera 1 names A within a few seconds of entering ("by door").
- ☐ A walks into camera 2's view with feet hidden behind a desk -> A's name appears on camera 2 (handoff), no wrong name on anyone for the whole walk. Note the delay: ___ s.
- ☐ A bends, crouches, sits, leans -> keeps the name (or shows "?"), never takes B's name.
- ☐ A and B cross twice, close together -> names stay on the right person both times.
- ☐ A walks behind the FANUC enclosure / entrance wall for ~4 s and comes back -> same name on return.
- ☐ Restart **one** camera service while A stands in view -> A first shows "A ? (look)" (guess), then the confirmed name when the face is matched. Note: ___ s to the guess, ___ s to the confirmed name.
- ☐ A visitor C (not enrolled, similar clothes to A) walks in -> C stays "Person N" / Unknown, never "A".
- ☐ Live bar under the cameras on the Security page lists every person with colour, posture and nearby equipment, and updates within ~2 s.
- ☐ Each camera keeps ~9 analyses/s with both running (Security page metrics). Measured: cam1 ___ / cam2 ___.

Portal

- ☐ Phone and laptop: switch between Dashboard, Labs, Devices, Bookings twice - the second visit shows content at once (no skeleton). Note any page that still blanks: ___
- ☐ Assistant: a free question shows the answer appearing word by word; "what is everyone doing right now" answers instantly with the camera lines.
- ☐ Sign out and sign in as a student: none of the staff data is shown, even for a moment.

Phone AR ("Scan the lab")

- ☐ Two lab PCs in one frame -> two boxes. Router, switch, master node in one frame -> each boxed once.
- ☐ Hold the phone still 10 s -> boxes do not jump. Note the time from pointing to first box: ___ s.

Devices and security

- ☐ Door, environment, consumables, reusables nodes still report (Devices page: all online).
- ☐ Installing a new lab PC: set `PC_ENROLL_OPEN=true` in `.env`, restart the backend, install, then set it back to false (enrolment is now closed by default).

What to bring back for me (so the remaining tuning is based on real data)

Everything stays on the laptop; nothing is uploaded. Ask the people filmed first.

1. **Two synchronized camera recordings, ~10 min, two enrolled people.** Portal -> Security -> enlarge a camera -> Record, on **both** cameras, then walk this script (one step at a time, say the step out loud for the audio): enter through the door one after the other (badge + fingerprint/face); walk the length of the lab; stand and sit at two PCs; bend down to a bin; crouch; walk behind the FANUC cell and the entrance wall (~4 s hidden); cross paths twice close together; stand side by side 30 s; one person leaves the room for 1 min and comes back; finally restart one camera service (`stop_all.bat` is NOT needed - close only that camera's window and run launch.bat) while both stand in view. Files land in tools\security_vision\recordings\ (local). This is what I use to measure real re-ID distances, the half-body matching and the handoff delay, and to set thresholds on evidence.
2. **Seats:** sit alone at each PC both cameras see and run `py tools\security_vision\seat_capture.py PC-07` (then PC-08, ...). About 5 s each; fixes "PC-08 in use" when someone sits at PC-07.
3. **Floor pairs:** with one person standing still at 4-5 different spots seen by both cameras, run `py tools\security_vision\pair_capture.py "Ali at the UR5e"` at each spot. Improves the two-camera floor match.
4. **AR photos for the weak classes:** 30-50 phone photos each of the network switch, the FANUC robot, both teach pendants, lab PCs and the door, from different distances and angles, plus 20 photos with several of them in one frame. A 30 s screen recording of "Scan the lab" moving across the room. Put them in a folder; I add them to the Colab round.
5. **The numbers in the checklist above** (delays, analysis rates, anything that went wrong with the time it happened).
6. **Logs:** nothing to do - the services write them to `logs\` and `tools\security_vision\recordings\identity_log*.jsonl`.

Not to be done as part of this checklist

Door relay tests, firmware flashing, robot motion, changing the admin password.